# MZUZU UNIVERSITY
## Department of Mathematics

## EMM 1101: Precalculus
*End of Semester Examination*

**20**$^{th}$ *April*, 2017 *Time* : 08 : 00 − 11 : 00 (3*hours*)

(a) Attempt **all** questions.

(b) Simplify your answers as far as possible.

(c) Do not use Calculators.

1. Use properties of exponents to simplify

   (a) $\left(\dfrac{x^2y^{20}}{x^6y^{15}}\right)^2$ (2)

   (b) $(3^{1+\sqrt{5}})(3^{1-\sqrt{5}})$ (2)

2. Simplify the expression $\sqrt[5]{\dfrac{16a^{\frac{1}{3}}}{a^{\frac{1}{4}}}}$. (3)

3. Simplify $|x+1|+4|x+3|$ given that $x<-3$. (3)

4. Show that $\triangle ABC$ with vertices $A(-7,4), B(-3,10), C(1,3)$ is an isosceles triangle. (4)

5. Sketch a graph of the function $A(x)$ and state the domain and range (5)

$$A(x) = \begin{cases} x^3, & -2 \le x \le -1 \\ x^2, & x > -1 \end{cases}$$

6. The center of a circle is the point $(3,2)$. If the point $(-2,-10)$ lies on this circle, find the standard equation of the circle . (5)

7. (a) Find the value of the constant $k$ for which $x^2+2(k+1)x+k^2=0$ has equal roots. (4)

   (b) Solve the equation $2x^2=3-4x$ and leave your answer in irrational form. (6)

8. Solve the inequality $\left|\frac{3x}{5}-\frac{x-1}{3}\right|<1$ and express your answer in interval notation. (4)

9. Let $f(x)=\frac{2x-3}{x+4}$.

   (a) State the domain and range of the function $f(x)$ (3)

   (b) Find $f^{-1}(x)$ (3)

10. (a) Use translations and reflections to graph the function (4)

$$y=-3x^2+12x$$

    (b) Use your graph to specify the vertex (turning point), axis of symmetry, maximum or minimum value and intercepts of $y=-3x^2+12x$. (4)

11. (a) Write the equation $\sqrt{2}=2^{\frac{1}{2}}$ in logarithmic form. (2)

    (b) Evaluate $\log_4 \frac{1}{16}$. (3)

12. Suppose that $\cos\theta=-\frac{1}{3}$, $(\frac{\pi}{2}<\theta<\pi)$, compute $\cot\theta$ (3)

13. Evaluate each expression

    (a) $\cos(-135)$ (2)

    (b) $\tan\frac{11\pi}{6}$ (2)

14. (a) Convert $330^\circ$ to radian measure. (2)

    (b) Show that $\sin^2 t-\cos^2 t=\frac{1-\cot^2 t}{1+\cot^2 t}$ (4)

15. Sketch the graph of $y=\ln(x+2)$ and state the asymptote, $y$ and $x$ intercepts. (5)

16. Solve for $x$: $\log_3 x+\log_3(x+2)=1$ and write your answer using interval notation. (4)